This roadmap is very close to what is followed by AI engineers in industry, but I would reorganize a few topics to make the learning progression smoother. For example, **Git, Linux, APIs, networking, software engineering, cloud basics, and system design** are essential skills that AI engineers use every day, so they should be included before the advanced AI topics.

Below is a **complete AI Engineering Curriculum (Beginner → Expert)** with detailed topics for every phase.

🚀 Phase 0 – Computer Science Fundamentals

Module 0.1 - Introduction to Computing

- What is a Computer?
- History of Computers
- Types of Computers
- Hardware vs Software
- Input → Process → Output
- Computer Architecture

Module 0.2 - Operating Systems

- What is an Operating System?
- Windows
- Linux
- macOS
- File System
- Processes
- Threads
- Memory Management

Module 0.3 - Computer Hardware

- CPU
- GPU
- RAM
- Storage (SSD/HDD)
- Cache Memory
- Motherboard

- Network Card

Module 0.4 - Binary & Data Representation

- Decimal
- Binary
- Octal
- Hexadecimal
- Binary Arithmetic
- Bits & Bytes
- Character Encoding (ASCII, Unicode, UTF-8)

Module 0.5 - Networking Fundamentals

- Internet
- LAN
- WAN
- IP Address
- DNS
- HTTP
- HTTPS
- TCP/IP
- Client-Server Architecture
- REST APIs
- JSON
- WebSockets

Module 0.6 - Software Engineering Basics

- SDLC
- Agile
- Scrum
- Git
- GitHub
- Version Control
- CI/CD Basics

Module 0.7 - Linux Fundamentals

- Terminal

- File Commands
- Permissions
- Shell
- Environment Variables
- Bash Scripting

Module 0.8 - Problem Solving

- Flowcharts
- Pseudocode
- Computational Thinking
- Algorithm Thinking

🐍 Phase 1 – Python Programming

Basics

- Installation
- IDE Setup
- Variables
- Data Types
- Operators
- Input Output
- Type Casting

Control Flow

- if
- elif
- else
- match
- Nested Conditions

Loops

- for
- while
- break
- continue

- pass

Functions

- Parameters
- Arguments
- Return
- Scope
- Lambda
- Recursion
- Decorators
- Generators

Data Structures

- List
- Tuple
- Set
- Dictionary
- String Operations

OOP

- Class
- Object
- Constructor
- Inheritance
- Encapsulation
- Polymorphism
- Abstraction
- Magic Methods

Advanced Python

- Modules
- Packages
- Virtual Environment
- Exception Handling
- File Handling
- Logging

- JSON
- CSV
- XML
- Datetime
- Collections
- itertools
- functools
- typing

📚 Phase 2 – Data Structures & Algorithms

Data Structures

- Arrays
- Linked Lists
- Stack
- Queue
- Deque
- Hash Table
- Trees
- Binary Tree
- BST
- Heap
- Trie
- Graph

Algorithms

- Time Complexity
- Space Complexity
- Big O
- Searching
- Sorting
- Binary Search
- DFS
- BFS
- Recursion
- Divide & Conquer
- Greedy
- Dynamic Programming
- Backtracking

➗ Phase 3 – Mathematics for AI

Linear Algebra

- Scalars
- Vectors
- Matrices
- Matrix Multiplication
- Determinants
- Inverse
- Rank
- Eigenvalues
- Eigenvectors
- SVD

Calculus

- Functions
- Limits
- Derivatives
- Partial Derivatives
- Gradients
- Chain Rule
- Optimization
- Gradient Descent

Optimization

- Convex Functions
- Cost Functions
- Loss Functions
- Learning Rate

📊 Phase 4 – Statistics & Probability

Statistics

- Mean
- Median

- Mode
- Variance
- Standard Deviation
- Percentiles
- Correlation
- Covariance
- Sampling
- Confidence Intervals
- Hypothesis Testing

Probability

- Sample Space
- Events
- Conditional Probability
- Bayes Theorem
- Random Variables
- Probability Distributions
- Normal Distribution
- Binomial Distribution
- Poisson Distribution

Phase 5 – SQL & Databases

SQL

- SELECT
- INSERT
- UPDATE
- DELETE
- WHERE
- GROUP BY
- ORDER BY
- HAVING
- JOIN
- UNION
- Views
- Indexes
- Stored Procedures
- Transactions

Database Design

- ER Diagram
- Normalization
- Relationships
- Constraints
- ACID Properties

NoSQL

- MongoDB
- Document Databases
- Key-Value Databases
- Graph Databases

⚙️ Phase 6 – Data Engineering Basics

- ETL
- ELT
- Data Warehouse
- Data Lake
- Batch Processing
- Stream Processing
- Data Pipelines
- Apache Spark Basics
- Kafka Basics
- Airflow Basics
- Parquet
- ORC
- Avro

🐼 Phase 7 – Python Libraries for Data Science

NumPy

- Arrays
- Broadcasting
- Vectorization

- Linear Algebra
- Random Module

Pandas

- DataFrames
- Series
- Indexing
- Filtering
- GroupBy
- Merge
- Pivot
- Window Functions

Visualization

- Matplotlib
- Plotly

Scientific Libraries

- SciPy
- Statsmodels

🧹 Phase 8 – Data Cleaning & Preprocessing

- Missing Values
- Duplicate Removal
- Outlier Detection
- Encoding
- Scaling
- Normalization
- Standardization
- Feature Engineering
- Data Validation
- Data Transformation
- Data Leakage

- Imbalanced Data
- SMOTE

📈 Phase 9 – Exploratory Data Analysis (EDA)

- Univariate Analysis
- Bivariate Analysis
- Multivariate Analysis
- Correlation Matrix
- Distribution Analysis
- Feature Relationships
- Business Insights
- Data Visualization
- Dashboard Creation

🤖 Phase 10 – Machine Learning

Fundamentals

- ML Workflow
- Types of ML
- Model Lifecycle

Supervised Learning

- Linear Regression
- Logistic Regression
- KNN
- Decision Tree
- Random Forest
- SVM
- Naive Bayes
- XGBoost
- LightGBM
- CatBoost

Unsupervised Learning

- K-Means
- Hierarchical Clustering
- DBSCAN
- PCA
- t-SNE
- UMAP

Model Evaluation

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

Advanced

- Cross Validation
- Hyperparameter Tuning
- Grid Search
- Random Search
- Bayesian Optimization
- Ensemble Learning
- Model Explainability (SHAP, LIME)

🧠 Phase 11 – Deep Learning

- Biological Neuron
- Artificial Neuron
- Perceptron
- Neural Networks
- Activation Functions
- Loss Functions
- Optimizers
- Backpropagation
- CNN
- RNN
- LSTM

- GRU
- Autoencoders
- GANs
- Transfer Learning
- PyTorch
- TensorFlow
- Keras

💬 Phase 12 – Natural Language Processing (NLP)

- Text Cleaning
- Regular Expressions
- Tokenization
- Stemming
- Lemmatization
- Stop Words
- POS Tagging
- Named Entity Recognition
- Dependency Parsing
- Bag of Words
- TF-IDF
- Word2Vec
- GloVe
- FastText
- Text Classification
- Sentiment Analysis
- Topic Modeling
- Sequence Models
- Attention Mechanism
- Transformers
- Hugging Face
- Fine-Tuning NLP Models

🧠 Phase 13 – Large Language Models (LLMs)

- Transformer Architecture
- Self-Attention
- Multi-Head Attention
- Positional Encoding

- Encoder vs Decoder
- GPT Family
- BERT Family
- Llama
- Mistral
- Gemma
- Tokenization
- Context Window
- Prompt Engineering
- Chain of Thought
- Temperature
- Top-k
- Top-p
- Fine-Tuning
- LoRA
- QLoRA
- Quantization
- RLHF
- Model Evaluation

🔍 Phase 14 – Embeddings & Semantic Search

- Embeddings
- Sentence Embeddings
- Similarity Metrics
- Cosine Similarity
- Dot Product
- Euclidean Distance
- Embedding Models
- Semantic Search
- Re-ranking
- Hybrid Search

🗃️ Phase 15 – Vector Databases

- Vector Indexing
- FAISS
- Chroma
- Pinecone
- Weaviate
- Milvus

- Qdrant
- Metadata Filtering
- ANN Algorithms
- HNSW
- IVF
- Product Quantization

📚 Phase 16 – Retrieval-Augmented Generation (RAG)

- RAG Architecture
- Document Loaders
- Chunking
- Embedding Pipelines
- Retrieval
- Hybrid Retrieval
- Query Rewriting
- Re-ranking
- Prompt Templates
- Context Management
- Hallucination Reduction
- RAG Evaluation
- Graph RAG
- Multimodal RAG

🤖 Phase 17 – AI Agents

- Agent Architecture
- Reasoning
- Planning
- Memory
- Tool Calling
- Function Calling
- MCP (Model Context Protocol)
- Workflow Orchestration
- Reflection
- ReAct Pattern
- Agent Evaluation

Phase 18 – Agentic AI

- Agentic Workflows
- LangGraph
- CrewAI
- OpenAI Agents SDK
- Google ADK
- State Management
- Human-in-the-Loop
- Long-Term Memory
- Multi-Step Reasoning
- Error Recovery
- Workflow Automation

Phase 19 – Multi-Agent Systems

- Agent Roles
- Planner Agent
- Research Agent
- Coding Agent
- Reviewer Agent
- Communication Protocols
- Shared Memory
- Conflict Resolution
- Swarm Intelligence
- Distributed Agent Systems

Phase 20 – MLOps & LLMOps

- Docker
- Kubernetes
- FastAPI
- Model Serving
- MLflow
- DVC
- Weights & Biases
- CI/CD for ML
- Feature Stores
- Monitoring
- Drift Detection
- Prompt Versioning
- Model Registry

- Security & Guardrails
- Cost Optimization

🏭 Phase 21 – Production AI Applications

Backend

- FastAPI
- Flask
- Django
- REST APIs
- WebSockets

Frontend

- React
- Next.js
- Streamlit
- Gradio

Cloud & Deployment

- AWS
- Azure
- Google Cloud
- Docker Compose
- Kubernetes
- Nginx
- GitHub Actions

Capstone Projects

- AI Chatbot
- Document Q&A (RAG)
- AI Coding Assistant
- AI Email Automation
- Resume Analyzer
- Voice Assistant
- Multimodal AI Assistant
- AI Research Agent
- Customer Support Agent
- Autonomous Multi-Agent Platform

🎯 What we'll do for every topic

Every concept in every phase will be taught using the same structured approach:

1. **What is it?** (Simple explanation)
2. **Why do we need it?**
3. **Real-world analogy**
4. **How a computer understands it**
5. **Mathematical intuition** (where applicable)
6. **Programmatic implementation from scratch**
7. **Implementation using industry-standard Python libraries**
8. **Visual diagrams and workflows**
9. **Hands-on exercises**
10. **Mini project**
11. **Interview questions**
12. **Common mistakes and best practices**
13. **How it's used in real AI systems**
14. **Assignment and quiz**

This approach will take you from foundational computer science all the way to designing and deploying production-grade AI systems with a deep understanding of *why* each technology works, not just *how* to use it.